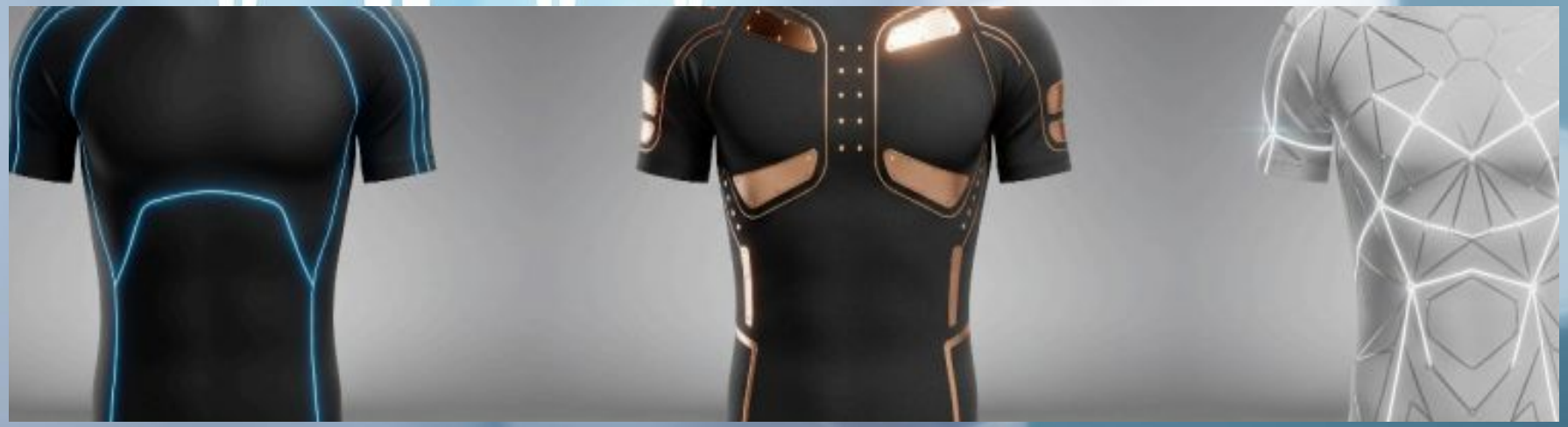
DRONACHARYA
College of Engineering

DEPARTMENT OF CSE(AI&ML)

Project II Presentation
NIVARAN: Enhancing Lives Through Wearable Health Tech

(Non-Invasive Vital and Risk Assessment through Wearable AI for Relief and Assistance in Nursing)



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Chapter 1: Introduction of Project

1.1 Introduction

NIVARAN is a wearable health-monitoring system designed to detect **stress**, **abnormal vitals**, and **falls** using **Machine Learning and a Web Dashboard**. The goal is to create a **non-invasive**, **low-cost**, and **continuous health-monitoring solution** for students, elderly people, and individuals recovering from medical conditions.Traditional devices like smartwatches are expensive and limited, so NIVARAN provides a **research-friendly, AI-based prototype** that integrates multi-sensor data, real-time analysis, and user-friendly visualization.

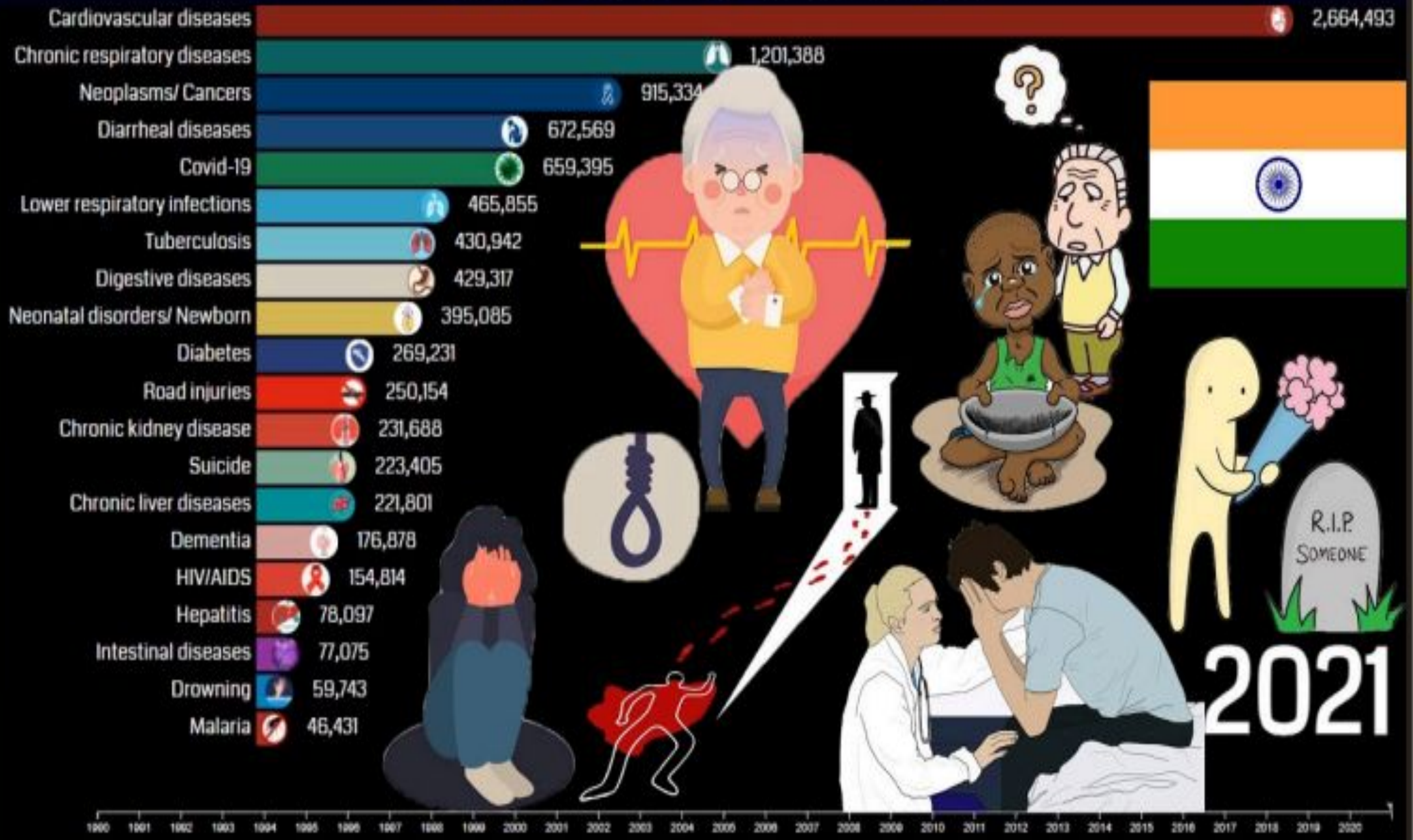
1.2 Project Overview

Background:
There is a lack of **affordable** and **continuous** health monitoring systems that can detect stress patterns, vital abnormalities, and accidental falls in real time. Existing solutions do not provide **AI-driven multi-sensor analysis** or **custom dashboards** accessible to caregivers.

Objectives:

- Build ML models for **stress** and **fall detection**.
- Develop a **web dashboard prototype** to visualize vitals.
- Create **flowchart + system architecture** for research & future patenting.
- Begin **wearable prototype design (3D model)** — hardware continues next semester.
- Enable early health risk alerts through AI-driven predictions.

Cause Of Death in India



Chapter 2: Tools & Technologies Used

Software Tools

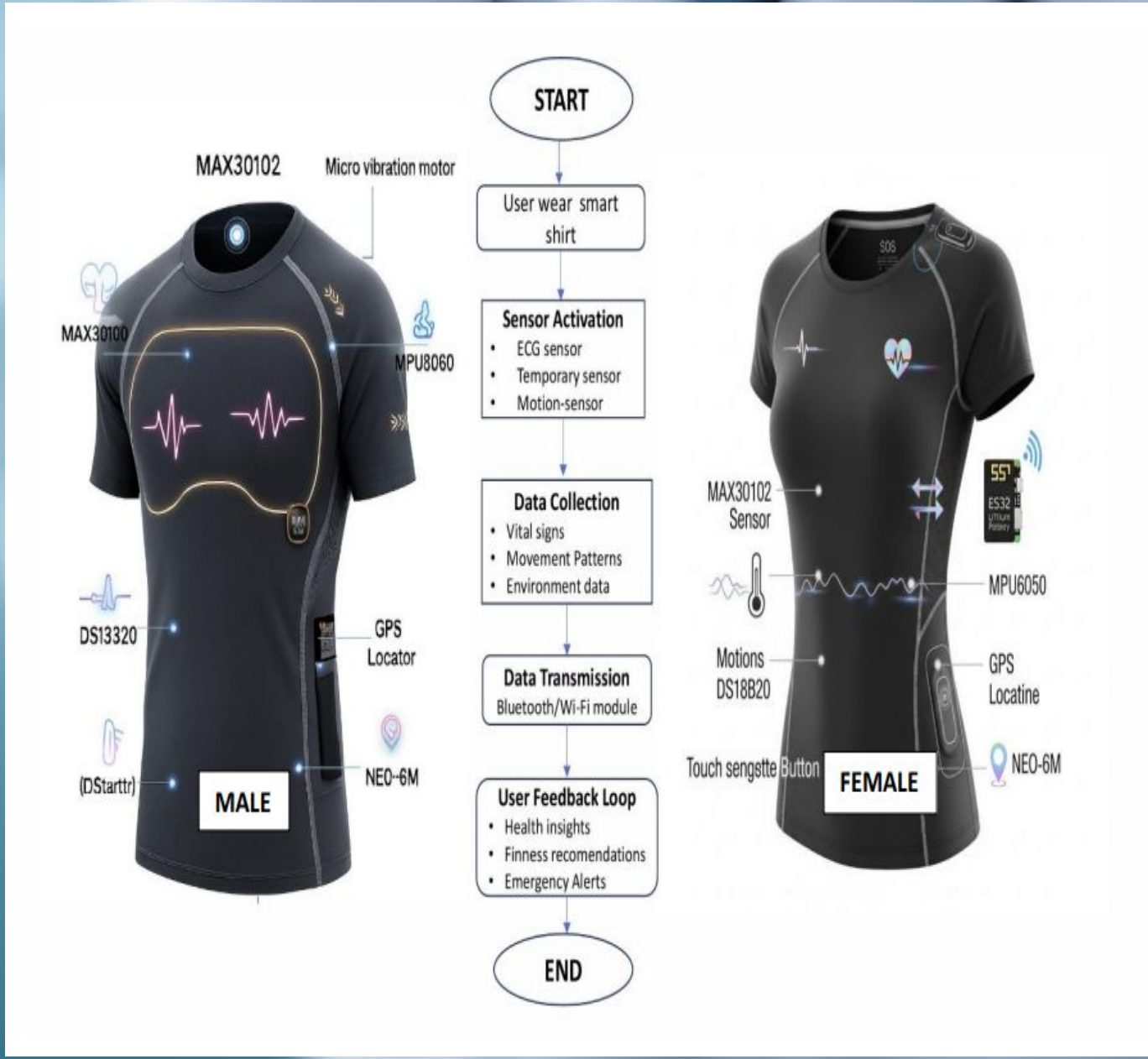
- Python, NumPy, Pandas, Scikit-learn
- Matplotlib, Seaborn
- React.js dashboard (prototype)
- Joblib for model deployment

Hardware (For Phase-2 Patent Prototype)

- ESP32
- MAX30102 (HR & SpO_2)
- DS18B20 (Temperature)
- MPU6050 (Motion)
- GPS Module
- Vibration Motor

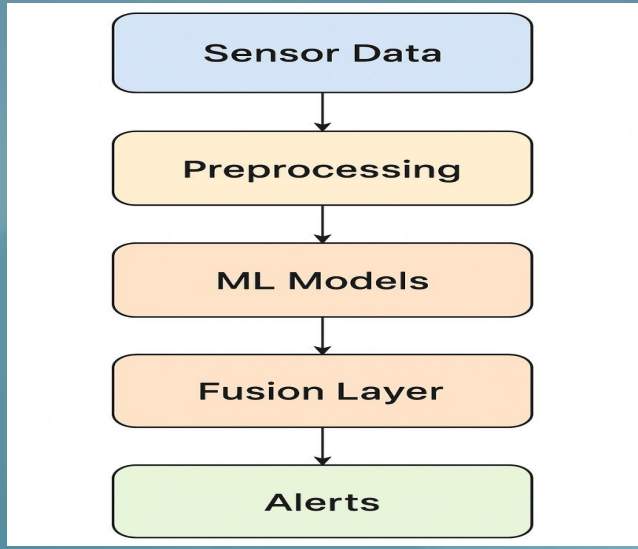
Research & Security Tools

- Blockchain-based logging (planned)
- Supabase for secure user authentication (planned)
- FastAPI backend (planned)



Chapter 3: Flowchart, Methodology & Code Outputs

3.1 Flowchart



3.2 Methodology

- **Model A (Stress & Vitals):** Random Forest trained on *wearable_sensor_data.csv*

```
✅ Model A Accuracy: 100.00%
💾 Saved models/modelA_stress_vitals_rf.pkl
```

- **Model B (Fall Detection):** Random Forest trained on accelerometer dataset

```
✅ Model B Accuracy: 95.55%
💾 Saved models/modelB_fall_mobility_rf.pkl
```

- **Fusion Model:** Combines A & B probabilities for final classification
  - Accuracy: **100%**

```
💾 Saved models/fusion_model_probs_rf.pkl
```

3.3 Code Execution

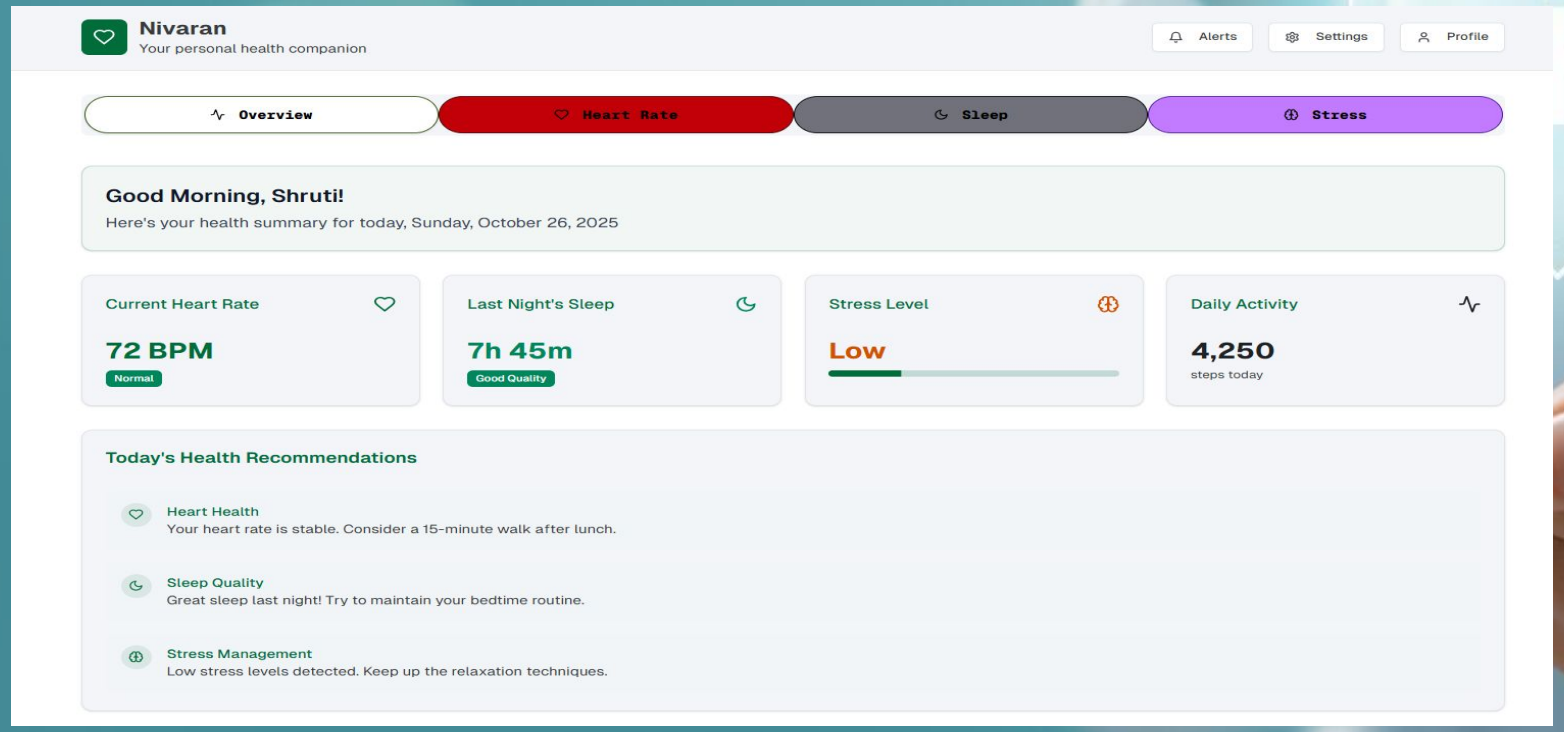
```
Enter values: 80,18,36.8,1,0,0

🩺Model A (Stress/Vitals):
  Input: HR=80, RR=18, Temp=36.8, Motion=1, Activity=0
  Predicted Class: 0, Probabilities: [[1. 0. 0.]]

📉 Model B (Fall Detection):
  Simulated Fall Status=0, Probs: [[0.9        0.01666667 0.01666667 0.01666667 0.01666667 0.01666667
  0.01666667]]

🔗 Fusion Model Result:
✅ Status: Normal.
```

Chapter 4: Implementation Timeline & Dashboard Prototype



Chapter 5: Conclusion & Future Scope

Conclusion

This project successfully demonstrates the foundation for an AI-based smart health monitoring system.
We completed the **machine learning models**, **fusion pipeline**, **dashboard**, and a **3D prototype** within one month.

Even though our project was not selected in FAER, we extended the work independently and developed a strong **ML + Dashboard software prototype**, forming the foundation for future patent and research work.

Future Scope

- Integrate real sensor data with hardware T-shirt
- Complete Phase-2 functional prototype for **patent filing**
- Publish research work in IEEE/Springer conferences
- Implement FastAPI + Supabase authentication
- Add mobile app for live alerts
- Test prototype with real users and expand dataset

References

[1] F. Sanfilippo and K. Y. Pettersen, "A sensor fusion wearable health-monitoring system with haptic feedback," in 2015 11th International Conference on Innovations in Information Technology (IIT), 2015, pp. 262–266. doi: 10.1109/INNOVATIONS.2015.7381551.

[2] J. Pansiot, D. Stoyanov, D. McIlwraith, B. P. L. Lo, and G. Z. Yang, "Ambient and Wearable Sensor Fusion for Activity Recognition in Healthcare Monitoring Systems," in 4th International Workshop on Wearable and Implantable Body Sensor Networks (BSN 2007), 2007, pp. 208–212.

[3] X. Chai, R. Wu, M. Pike, H. Jin, W.-Y. Chung, and B.-G. Lee, "Smart wearables with sensor fusion for fall detection in firefighting," Sensors, vol. 21, no. 20, p. 6770, 2021.